

DESCRIPTIVE STATISTICS

Load the dataset 'k_circles_sales' as a csv file and read it using pandas

Get the basic information about data:

A) No. of rows and columns

B) Datatypes

C) Columns

D) Description

1 .Count values in item_fat_content column

2. Find unique values in item_type.

3. Replace LF with Low Fat and reg with regular in item_fat_content.

4. Find the mean weight of all soft drinks

5. Find mean weight of items whose profit is more than 10.5

6. Find IQR of item_MRP

7. Find the 30th percentile of item_MRP.

8. Find variance, standard deviation, coefficient of variation of item_MRP

9. Find skewness and kurtosis of item_weight and profit.

10. Prepare a histogram for item_type.

To replace the values

```
10.df.replace(to_replace="LF", value="Low Fat", inplace=True)
11.df_sales.replace(to_replace="reg", value="Regular", inplace=True)
```

To find mean

```
13.df[df['Item_Type'] == 'Soft Drinks']['Item_Weight'].mean()
14.df[df['profit']>10.5]['item_weight'].mean()
```

To find median and mode

```
16.df.select_dtypes(include="number").median()
17.df.median(numeric_only=True)
```

```
18.df.select_dtypes(include=object).mode()
```

```
26.## FILL MISSING VALUES WITH MEAN (numeric only)
```

```
df[col]=df[col].fillna(df[col].mean)
```

To find quantiles

```
|
23.df.quantile(0.25,numeric_only=True)  (Q1)
24.df.quantile(0.2,numeric_only=True)   (D2)
25.df.quantile(0.20,numeric_only=True)  (P20)
    IQR= df.quantile(0.75,numeric_only=True)-df.quantile(0.25,numeric_only=True)
```

```
28.##variance
df['item_weight'].var()
df.var(numeric_only=True)
```

```
29.## standard deviation
df['item_weight'].std()
df.std(numeric_only=True)
```

```
30.coefficient of variation
df['item_weight'].variation()
31.df['item_weight'].skew()
    df.skew(numeric_only=True)
32.df['item_weight'].kurt()
    df.kurt(numeric_only)
```

```
33.## for creating normal distribution
data= np.random.normal(50,10,50)
mean = np.mean(data)
median = np.median(data)
mode = stats.mode(data)[0]
```

```
34. ## for creating skewed data
data1= np.random.normal(50,10,50)**5
mean = np.mean(data1)
median = np.median(data1)
mode = stats.mode(data1)[0]
```